Longitudinal Analysis of Motor Subtype Transitions in Parkinson's Disease

Weeks 3–4 Progress Report

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Abstract

This report covers Weeks 3–4 of the project: tracking TD/PIGD motor subtype changes across visits, estimating transition probabilities over different follow-up intervals, and visualising progression pathways. The analysis uses 3,361 consecutive labelled visit pairs from 934 sporadic PD patients in the PPMI cohort.

Four results shape the modelling that follows. *Indeterminate* is not a stable state but a crossing — only 19.2% of patients remain in it at the next visit. *PIGD is not absorbing*: 29.8% of PIGD visits are followed by a non-PIGD assessment. *35.2% of all observed subtype changes reverse at the very next visit*, so a transition matrix built from raw labels measures assessment noise as much as disease progression. Yet the movement is not symmetric: TD → PIGD outnumbers PIGD → TD by 253 to 141 ($p = 1.8 \times 10^{-8}$), so a genuine directional drift exists beneath the noise. Finally, transition probabilities scale strongly with the gap between visits, which means a single pooled transition matrix is not a well-defined quantity and the Week 5–7 model should work in continuous time.

1 Objective and scope

The project proposal assigns Weeks 3–4 the following tasks:

- Track subtype changes for each patient across visits.
- Compute transition frequencies (TD → TD, TD → Indeterminate, TD → PIGD, and so on).
- Estimate transition probabilities over different follow-up intervals.
- Visualise common progression pathways.

with the deliverable stated as *transition matrices and visualisations of subtype progression.* All four are complete. This report also records two analyses that were not requested but that materially change how the transition matrix should be read — a reversion analysis and a sustained-conversion analysis — because without them the headline numbers are misleading.

Everything here is descriptive. No model is fitted; that is Weeks 5–7.

2 Data

2.1 The panel

The input is `outputs/visit_panel.csv`, built in Weeks 1–2 from the PPMI Curated Data Cut joined to the item-level MDS-UPDRS Part II and Part III files. Subtype labels follow Stebbins et

al. (2013):

$$\text{ratio} = \frac{\text{mean of 11 tremor items}}{\text{mean of 5 gait/postural items}}$$

with ratio $\geq 1.15 \Rightarrow$ TD, ratio $\leq 0.90 \Rightarrow$ PIGD, and the interval between them Indeterminate. Scores are taken from the OFF-medication or untreated examination.

Quantity	**Value**
Patients (sporadic PD)	1,080
Visits	5,774
Visits with a subtype label	4,441 (76.9%)
Patients contributing ≥ 1 transition	934
Consecutive labelled visit pairs	3,361
Median interval between labelled visits	1.00 years
Maximum follow-up	14 years

Table 1: Analysis population.

2.2 Why the labels can be trusted

The recomputed labels were validated against both of PPMI's own derivations before any of this analysis was run:

Reference	**Classes**	**Agreement**
Curated cut `td_pigd`	2 (TD / non-TD)	4,431 / 4,431 = 100.00%
Archived `td_pigd_old`	3 (TD / Indet. / PIGD)	1,595 / 1,597 = 99.87%

Table 2: Label validation. The three-class check covers 390 patients and is the only available test of the Indeterminate/PIGD boundary, since the curated cut collapses those two classes.

2.3 Baseline distribution

At baseline: TD 71.7%, Indeterminate 10.1%, PIGD 18.1%. Pooled over all labelled visits: TD 65.1%, Indeterminate 9.9%, PIGD 25.0%.

3 Methods

3.1 Transition pairs

For each patient, visits with a non-missing label are sorted by elapsed time and adjacent pairs are formed. Two decisions matter:

1. **Unlabelled visits are skipped, not treated as terminal.** If a patient is labelled at years 0 and 2 but not at year 1, that yields one pair with a two-year interval rather than truncating the sequence.
2. **The elapsed interval is carried through.** PPMI visits are annual early on and biennial later, and patients miss visits. A pair is therefore not a fixed unit of time.

Time is measured from the patient's own first visit using `INFODT`, which PPMI releases at month resolution.

3.2 Transition matrix and uncertainty

The matrix is row-stochastic: entry (i, j) is the proportion of visits in state i followed by state j. Row counts differ by an order of magnitude (2,261 from TD against 334 from Indeterminate), so every probability is reported with a **Wilson score interval** rather than a point estimate alone. Wilson is used in preference to the normal approximation because the latter produces intervals that leave $[0, 1]$ for probabilities near the boundary, which several of these are.

3.3 Interval stratification

Pairs are binned by elapsed interval (≤ 1.5, 1.5–2.5, 2.5–4.0, > 4.0 years) and a separate matrix estimated in each bin.

3.4 Reversion analysis

For every observed change of state $A \to B$ where a third labelled visit exists, we record whether that third visit returns to A. This separates two situations a transition matrix reports identically: a patient whose phenotype genuinely progressed, and a patient whose single noisy assessment crossed a threshold and crossed back.

3.5 Time to conversion

Time from baseline to a patient's first PIGD assessment is estimated by Kaplan–Meier with Greenwood standard errors, censoring at the last labelled visit. Patients already PIGD at baseline are excluded from the risk set. Two definitions of conversion are compared:

- **naive** — a single PIGD assessment;
- **sustained** — PIGD at two consecutive labelled visits.

3.6 Pathways

Each patient's label sequence is reduced by collapsing immediate repeats, so (TD, TD, PIGD, PIGD) becomes TD $\to$ PIGD. Without this, pathway frequencies would be dominated by how often a patient attended rather than by what happened to them.

4 Results

4.1 Pooled transition matrix

	State at next visit			
Current state	**TD**	**Indeterminate**	**PIGD**	**Row n**
TD	**0.808**	0.080	0.112	2,261
Indeterminate	0.440	**0.192**	0.368	334
PIGD	0.184	0.114	**0.702**	766

Table 3: Transition probabilities between consecutive labelled visits ($n = 3{,}361$ pairs). Counts: TD row 1827/181/253; Indeterminate row 147/64/123; PIGD row 141/87/538.

From	To	n from	n trans.	Prob.	Wilson 95% CI
TD	TD	2261	1827	0.808	[0.791, 0.824]
TD	Indeterminate	2261	181	0.080	[0.070, 0.092]
TD	PIGD	2261	253	0.112	[0.100, 0.126]
Indeterminate	TD	334	147	0.440	[0.388, 0.494]
Indeterminate	Indeterminate	334	64	0.192	[0.153, 0.237]
Indeterminate	PIGD	334	123	0.368	[0.318, 0.421]
PIGD	TD	766	141	0.184	[0.158, 0.213]
PIGD	Indeterminate	766	87	0.114	[0.093, 0.138]
PIGD	PIGD	766	538	0.702	[0.669, 0.734]

Table 4: Transition probabilities with Wilson score intervals.

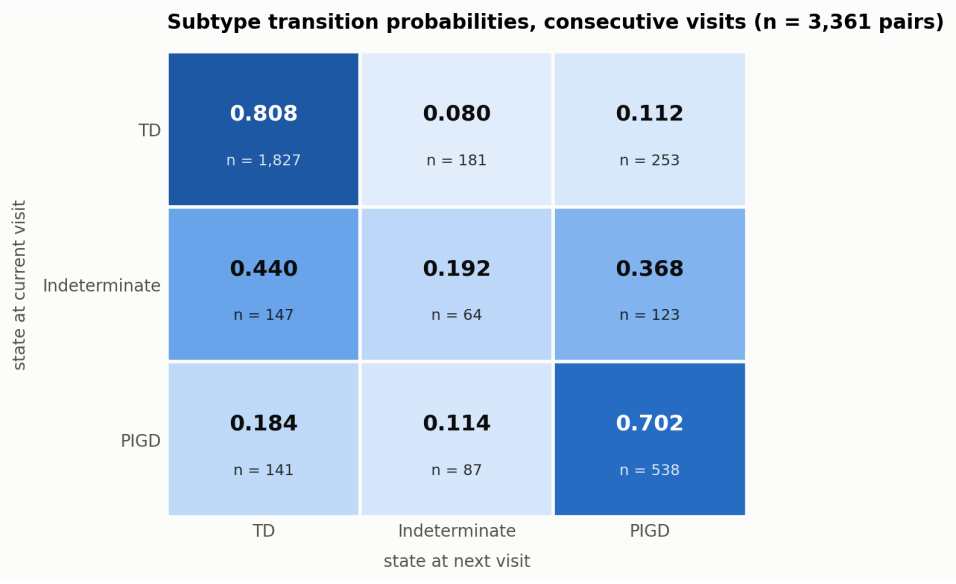


Figure 1: Pooled transition matrix. Colour encodes probability magnitude on a single-hue sequential ramp; cell counts are printed because the row totals differ by an order of magnitude.

4.2 Persistence of each state

From the diagonal, the expected number of consecutive visits spent in a state before leaving it is $1/(1 - p_{\text{stay}})$:

State	p_{stay}	Mean consecutive visits
TD	0.808	5.21
PIGD	0.702	3.36
Indeterminate	0.192	1.24

Table 5: State persistence. At a median interval of one year these are roughly years. Indeterminate barely outlives a single visit.

4.3 Directionality: is the movement symmetric?

High reversion could mean the labels are pure noise. Testing each pair of states against the null of symmetric exchange (exact binomial):

Pair	Forward	Reverse	Net	p
TD $\rightarrow$ PIGD	253	141	+112	1.8×10^{-8}
Indeterminate $\rightarrow$ PIGD	123	87	+36	0.016
TD $\rightarrow$ Indeterminate	181	147	+34	0.068

Table 6: Directional asymmetry. Of 932 cross-state moves, only 182 — 19.5% — are net drift; the remaining 80.5% cancels out. But the residue is real and, for TD $\leftrightarrow$ PIGD, overwhelmingly significant.

The single most important sentence in this report. Most observed subtype movement is bidirectional churn, but it does not cancel exactly: there is a statistically overwhelming net drift from TD toward PIGD. The data is therefore neither "clean progression" nor "pure noise" — it is a genuine drift observed through a noisy measurement. That is precisely the structure a latent-state model with a measurement-error term is built to represent, and it is the empirical justification for the proposed approach.

4.4 Dependence on the follow-up interval

	From TD			From Indeterminate			From PIGD		
Interval (yr)	TD	Ind	PIGD	TD	Ind	PIGD	TD	Ind	PIGD
≤ 1.5	0.820	0.079	0.100	0.477	0.197	0.326	0.192	0.115	0.693
1.5–2.5	0.762	0.069	0.168	0.200	0.200	0.600	0.100	0.100	0.800
2.5–4.0	0.750	0.097	0.153	0.375	0.125	0.500	0.174	0.130	0.696
> 4.0	0.578	0.133	0.289	0.222	0.111	0.667	0.250	0.083	0.667

Table 7: Transition probabilities stratified by elapsed interval. Row n, in the same interval order, is 1942/202/72/45 from TD, 279/30/16/9 from Indeterminate and 661/70/23/12 from PIGD. The Indeterminate and long-interval cells rest on very few observations and should be read as intervals, not point estimates.

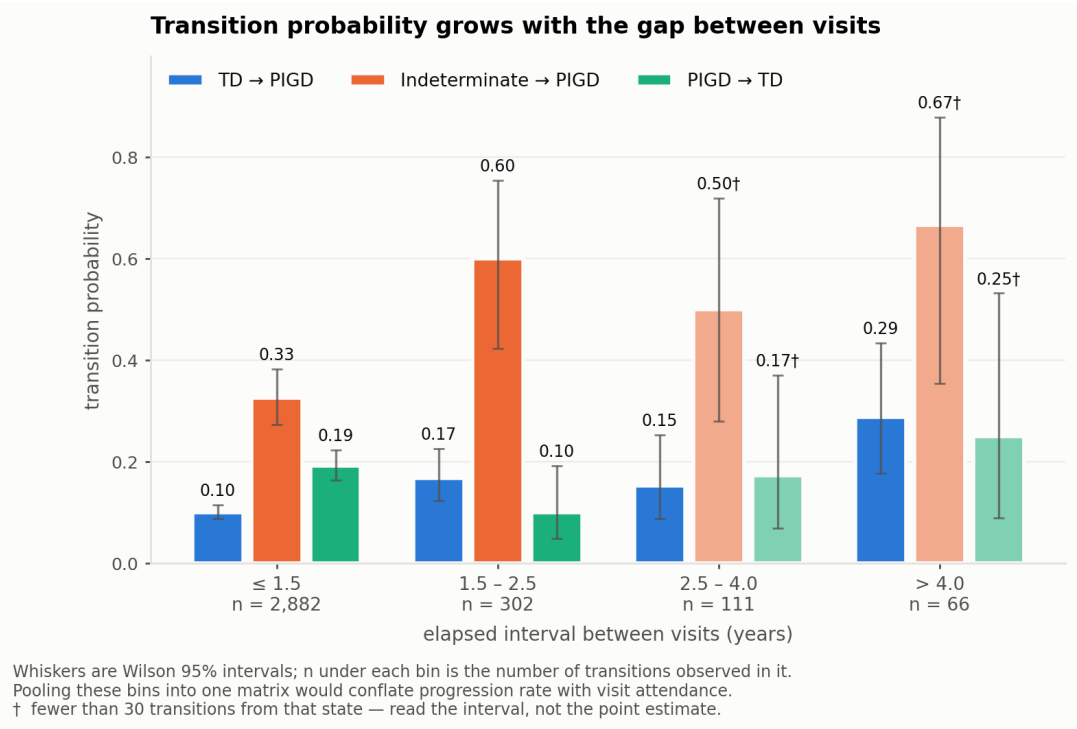


Figure 2: Selected transitions against elapsed interval. TD → PIGD nearly triples from 0.100 to 0.289 as the gap widens. Washed-out bars marked † rest on fewer than 30 transitions.

4.5 Reversion: how much of this is noise?

From	To	n changes	n reverted	Reversion rate
TD	Indeterminate	134	75	**0.560**
PIGD	Indeterminate	54	27	**0.500**
PIGD	TD	112	44	0.393
TD	PIGD	172	55	0.320
Indeterminate	PIGD	93	17	0.183
Indeterminate	TD	108	19	0.176
Overall		**673**	**237**	**0.352**

Table 8: Share of observed state changes that return to the origin state at the very next visit.

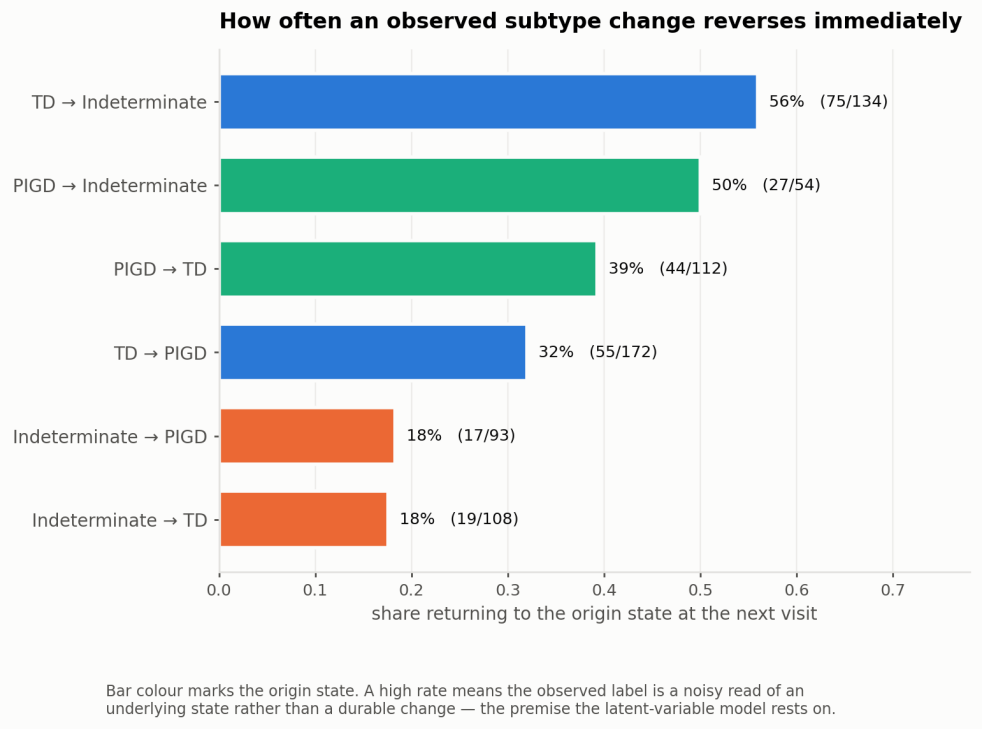


Figure 3: Reversion rates. Changes *into* Indeterminate reverse most often (50–56%), consistent with Indeterminate being a threshold band that a noisy ratio wanders across rather than a clinical state a patient occupies.

4.6 Time to PIGD conversion

	Naive (1 visit)	Sustained (2 visits)
At risk (not PIGD at baseline)	885	885
Converted	263 (29.7%)	127 (14.4%)
Median years to conversion	3.00	3.00
KM cumulative incidence, year 2	12.5%	6.5%
KM cumulative incidence, year 5	35.3%	15.3%

Table 9: Requiring the conversion to hold for two consecutive visits removes 136 of 263 apparent conversions — 51.7%.

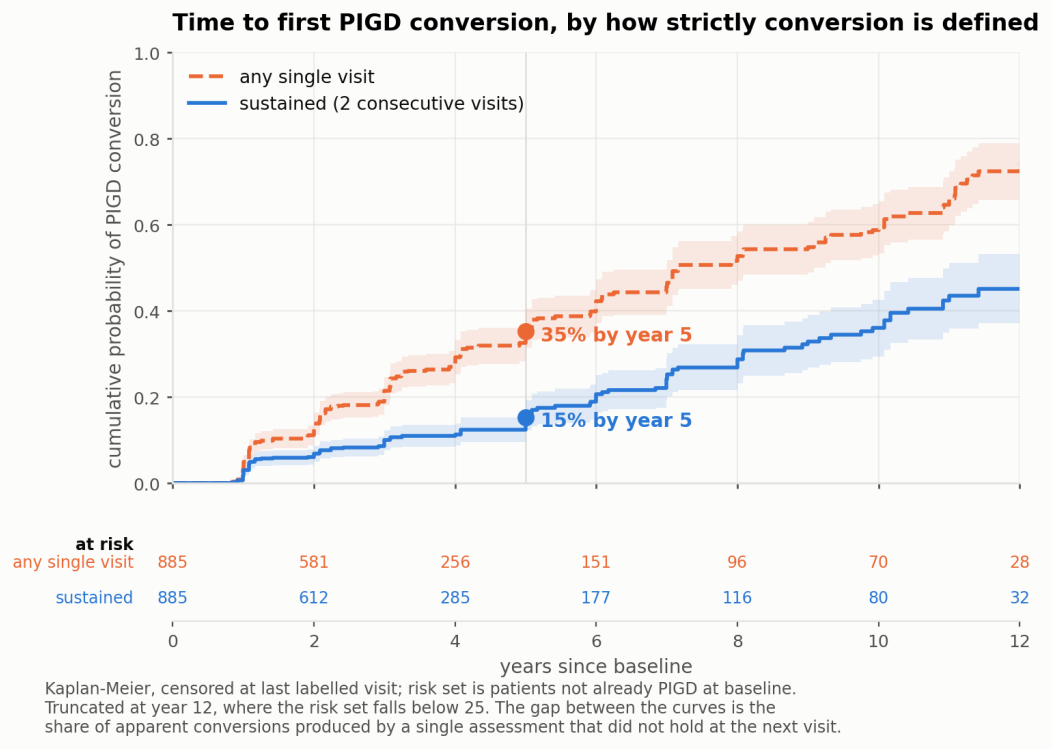


Figure 4: Cumulative incidence of PIGD conversion under the two definitions. Curves are truncated at year 12, where the risk set falls below 25; the at-risk counts are printed beneath the axis.

4.7 Prevalence across follow-up

Year	n TD	n Ind	n PIGD	% TD	% Ind	% PIGD	n total
0	772	109	195	71.7	10.1	18.1	1076
1	556	84	166	69.0	10.4	20.6	806
2	467	59	151	69.0	8.7	22.3	677
3	332	39	109	69.2	8.1	22.7	480
4	209	30	90	63.5	9.1	27.4	329
5	129	27	60	59.7	12.5	27.8	216
6	102	16	64	56.0	8.8	35.2	182
7	94	19	64	53.1	10.7	36.2	177
8	58	19	44	47.9	15.7	36.4	121
9	50	11	38	50.5	11.1	38.4	99
10	47	12	49	43.5	11.1	45.4	108
11	36	11	45	39.1	12.0	48.9	92

Table 10: Cross-sectional prevalence. Years 12–14 are omitted here ($n \leq 52$) but appear in `prevalence_by_year.csv`.

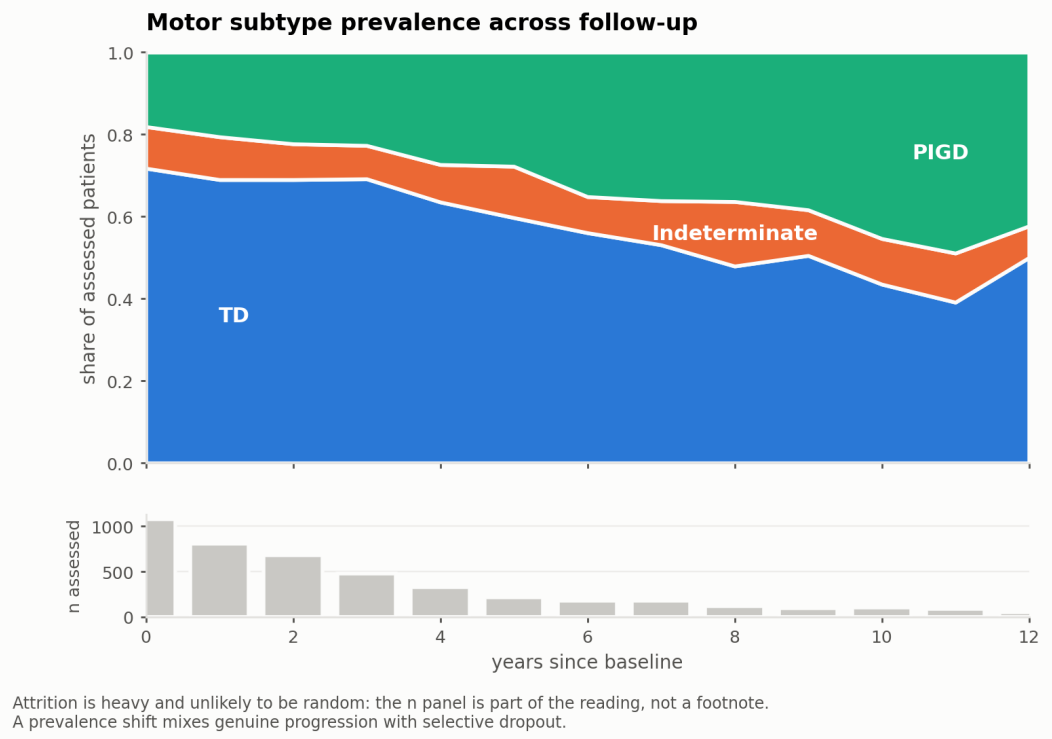


Figure 5: Subtype prevalence with the assessed-n panel beneath. The n panel is part of the reading: by year 8 fewer than 12% of the baseline cohort is still contributing.

4.8 Progression pathways

Pathway (repeats collapsed)	Patients	Share
TD	497	46.0%
PIGD	109	10.1%
TD → PIGD	83	7.7%
TD → Indeterminate	39	3.6%
TD → Indeterminate → TD	35	3.2%
Indeterminate → PIGD	27	2.5%
Indeterminate → TD	26	2.4%
PIGD → TD	24	2.2%
TD → Indeterminate → PIGD	21	1.9%
TD → PIGD → TD	20	1.9%
PIGD → TD → PIGD	19	1.8%
Indeterminate	16	1.5%

Table 11: The twelve commonest of 86 distinct pathways across 1,080 patients.

622 of 1,080 patients (57.6%) never change state at all. Median number of state changes per patient is 0; mean 0.86; maximum 10.

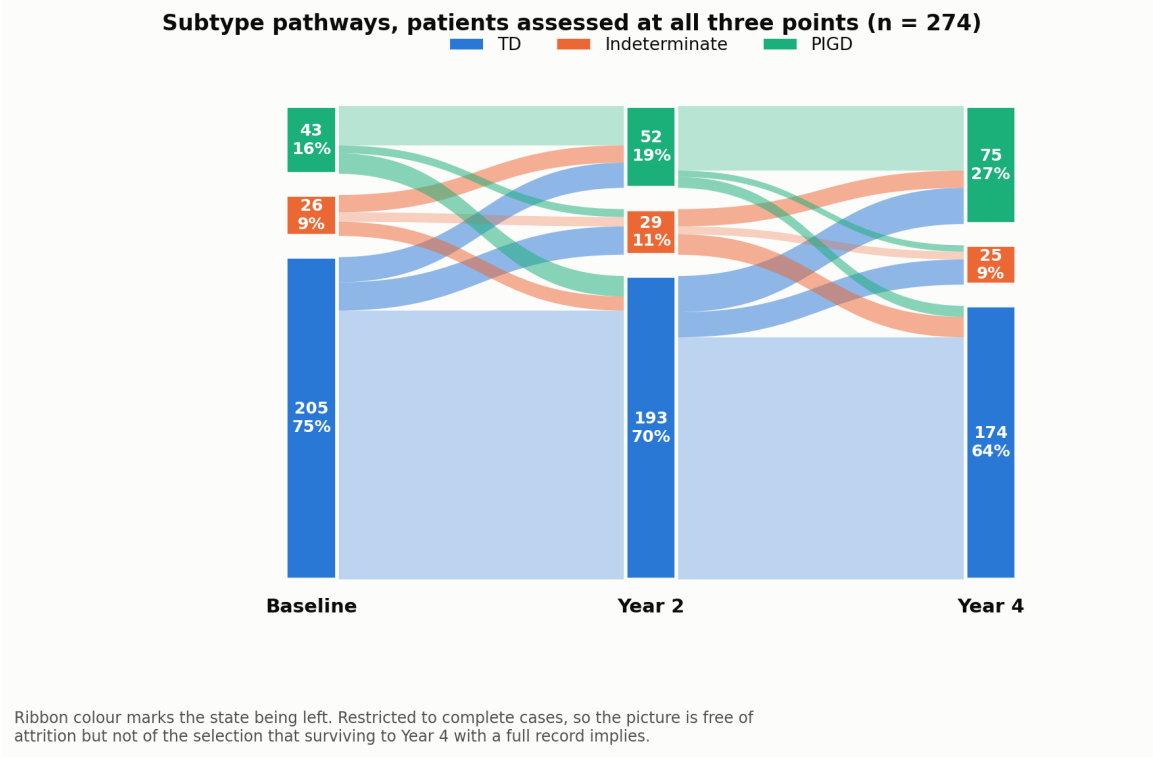


Figure 6: Subtype pathways for the 274 patients assessed at baseline, year 2 and year 4. Restricted to complete cases, so free of attrition but not of the selection that surviving with a full record implies.

5 What this means for the model

5.1 Four consequences

1. **Indeterminate should not be modelled as a clinical state.** With $p_{\text{stay}} = 0.192$ and reversion rates of 50–56% for changes into it, it behaves exactly as the neutral band of a continuous latent variable would: a region the ratio passes through, not a place a patient sits. This is direct empirical support for the proposal's design, which models the continuous ratio and derives the three classes by thresholding, rather than modelling three discrete states.

2. **The observation model needs a real noise term.** 35.2% immediate reversion cannot be explained by disease progression. The proposal's observed = latent + noise structure is not a modelling convenience here; it is required to fit what the data does.

3. **The model must work in continuous time.** TD → PIGD nearly triples between the shortest and longest interval bins. A discrete Markov chain over "consecutive visits" would absorb PPMI's attendance schedule into its progression estimate. The natural fix is a continuous-time formulation — estimate a generator matrix Q and use $P(t) = e^{Qt}$ — or, following the proposal, model the latent ratio as a function of actual elapsed time.

4. **Conversion must be defined before it is predicted.** The naive/sustained gap (35.3% versus 15.3% at five years) is larger than any effect the model is likely to estimate. This choice should be made and stated now, not discovered later.

5.2 A caution on the pooled matrix

Taking Table 3 at face value and iterating it gives a stationary distribution of TD 0.560, Indeterminate 0.103, PIGD 0.338. The observed year-11 prevalence is TD 0.391, Indeterminate 0.120, PIGD 0.489 — considerably more PIGD than the pooled matrix predicts. At least three things could explain the gap: transition rates are not constant over disease duration, the process is not Markov in the observed label, or attrition is selective. A fourth possibility is developed in Section 6.1 — that the transition matrix itself is estimated on a sample from which the most severely affected visits have been systematically dropped, so its rates understate the true drift. Disentangling these is a Week 5–7 question, but the discrepancy is worth stating rather than presenting a stationary distribution as a finding.

6 Limitations

6.1 Label availability is not random, and it is not neutral

Of the 5,774 cohort visits, 1,333 (23.1%) carry no subtype label. That missingness is not a random thinning of the data, and its direction matters.

Reason a visit has no label	n	Share of the loss
No OFF-medication Part III exam recorded	722	54.2%
OFF exam recorded but items incomplete	595	44.6%
Part II missing or incomplete	16	1.2%

Table 12: Why visits go unlabelled.

Label availability also falls sharply across follow-up: 99.6% of baseline visits are labelled, 75.3% at year 4, 55.4% at year 11, and 19.2% at year 13.

The second route is the concerning one. Among the 1,253 OFF-medication exams that were recorded but could not be scored, **NP3PSTBL — item 3.12, postural stability — is missing in 1,243 of them**, and 185 of those are explicitly coded `101`, "Unable to Rate". The pull test is frequently unsafe or impossible to administer in a patient with poor balance.

Because our rule requires all sixteen items, one unscored postural-stability test discards the whole visit. And the visits it discards are not a random sample: exams with an incomplete item set come from materially more advanced patients, with a median Hoehn & Yahr stage of 3.0 against 2.0 for complete exams (Mann–Whitney $p = 1.4 \times 10^{-17}$).

This biases the results in a knowable direction. Postural stability is one of the five *PIGD* items. The visits we lose are therefore disproportionately those of patients whose gait and balance are worst — the patients most likely to be classified PIGD. The observed drift toward PIGD, and the PIGD conversion rates in Table 9, are for that reason most likely *underestimates*. This also offers a plausible explanation for the gap noted in Section 5.2, where observed year-11 PIGD prevalence exceeds what the pooled matrix predicts.

Two things are worth noting about how this was established. First, an earlier reading of the same numbers attributed the loss mainly to ON-state exams replacing OFF-state ones as patients become medicated. That explanation accounts for only about half the loss and plateaus after year 5, whereas incomplete item sets persist and are the clearly severity-linked route; the severity comparison above is what distinguished them. Second, the effect is a consequence of the strict all-items rule inherited from PPMI's own derivation. Relaxing it to tolerate one missing item was tested in Weeks 1–2 and recovers only 44 visits while changing no existing label, so relaxation is not a remedy — the incomplete exams are typically missing many items, not one.

A sensitivity analysis that imputes the postural-stability item, or that models the label as missing-not-at-random, would quantify the size of this bias. That is a Weeks 5–7 task.

6.2 Other limitations

- **Sparse cells.** Transitions out of Indeterminate at intervals beyond 1.5 years rest on 9–30 observations. The Wilson intervals are reported for exactly this reason and those cells are flagged in the figure.
- **Selective attrition.** By year 8 fewer than 12% of the baseline cohort is still assessed. This is separate from, and compounds, the label-availability problem in Section 6.1: a visit must both happen and be scorable to contribute.
- **Medication state.** Labels use the OFF or untreated exam, which is correct, but the proportion of patients who are genuinely untreated falls over follow-up — from 99.7% at baseline to under 2% by year 7. Some of the observed churn may reflect changing treatment context rather than phenotype.
- **Month-resolution dates.** PPMI releases `INFODT` as month and year only, so intervals carry roughly ± two weeks of noise. Negligible at annual spacing.
- **The reversion analysis needs three visits.** Patients with exactly two labelled visits contribute to the transition matrix but not to Table 8, so the reversion estimate is drawn from the better-attended subset.
- **Ratio undefined for 16.3% of labelled visits** (725 of 4,441) where the PIGD subscore is exactly zero. Those visits are classifiable — the rule sends them to TD — but carry no continuous observation. This is carried over from Weeks 1–2 and remains open.
- **Sporadic PD only.** Genetic PD subgroups (LRRK2, GBA, and others) are excluded. Extending to them is a configuration change.

7 Anticipated questions

Q. If 35% of changes reverse immediately, are the labels just noise?
No. If the movement were pure noise it would be symmetric, and it is not: TD → PIGD outnumbers PIGD → TD 253 to 141, $p = 1.8 \times 10^{-8}$ (Table 6). Of 932 cross-state moves, 80.5% cancels and 19.5% is net drift toward PIGD. So there is a real signal, observed through a noisy measurement — which is the case for a latent-variable model, not against it.

Q. Why is Indeterminate so unstable? Is that a bug?
It is expected, and it was checked. Indeterminate is defined as $0.90 < \text{ratio} < 1.15$ — a narrow band on a continuous, noisily measured quantity. Anyone whose true ratio sits near the band will cross in and out between visits. The label logic was validated at 99.87% against PPMI's own three-class derivation, so the instability is a property of the definition, not an error in the implementation.

Q. Why not just report one transition matrix?
Because it is not a well-defined quantity. TD → PIGD is 0.100 at intervals under 1.5 years and 0.289 beyond 4 years. A pooled matrix averages these in whatever proportion PPMI's visit schedule and patient attendance happened to produce, so it describes the study design as much as the disease.

Q. Which conversion definition should we use?
The sustained one (two consecutive PIGD assessments) for anything presented as a clinical outcome, because half of single-visit conversions do not hold. The naive one is worth reporting alongside it to show the size of the effect. What must not happen is choosing silently.

Q. Why Kaplan–Meier rather than counting conversions?
Because follow-up length varies enormously — median 3 visits, maximum 14 years. A raw proportion would conflate "did not convert" with "was not observed long enough". KM censors correctly at the last labelled visit. The curves are truncated at year 12 where the risk set drops below 25, since beyond that each step is driven by a handful of patients.

Q. Why Wilson intervals?
Row counts range from 334 to 2,261, and several probabilities sit near 0 or 1. The normal approximation produces intervals that fall outside $[0, 1]$ in exactly those cells; Wilson does not.

Q. Is PIGD an absorbing state, as the literature often assumes?
Not in this data. 29.8% of PIGD visits are followed by a non-PIGD assessment, 18.4% of them straight back to TD. Some of that is measurement noise — the PIGD → TD reversion rate is 39.3% — but a model that forbids leaving PIGD would be misspecified.

Q. You lose 23% of visits. Does that break the analysis?
It does not break it, but it biases it, and in a direction we can name. Roughly half the loss is visits with no OFF-medication exam; the other half is exams that were performed but could not be fully scored — and in 1,243 of those 1,253 exams the missing item is postural stability, which is often unsafe to test in a patient with poor balance. Those exams come from more advanced patients (median Hoehn & Yahr 3.0 versus 2.0, $p = 1.4 \times 10^{-17}$), and postural stability is a PIGD item. So we preferentially lose the patients most likely to be PIGD, which makes the reported drift toward PIGD a lower bound. Section 6.1 sets this out in full.

Q. Then why not relax the all-items rule?
It was tested. Allowing one missing item per subscore recovers 44 visits and changes zero existing labels, because the incomplete exams are typically missing many items rather than one. The strict rule also matches PPMI's own derivation, which is what lets us validate at 100% against their `td_pigd`. The real remedy is imputation or an explicit missing-not-at-random model, in Weeks 5–7.

Q. What did you actually deliver this fortnight?
The four proposal tasks: subtype changes tracked per patient, transition frequencies computed, probabilities estimated across four interval strata, and pathways visualised. Plus two analyses that were not asked for — reversion and sustained conversion — which changed how the first four should be read. Outputs are 9 CSV tables, 6 figures and a text report, produced by one reproducible command, with 74 unit tests over the underlying code.

Q. What is next?
Weeks 5–7: the baseline probabilistic model. Given the interval dependence, a continuous-time Markov chain estimating a generator Q is a better baseline than the discrete chain originally sketched, and it sets up the hierarchical Bayesian stage directly.

8 Reproducing this report

```
python scripts/00_data_audit.py          # verify source files
python scripts/01_build_panel.py         # build labelled visit panel
python scripts/02_transition_analysis.py # this report's numbers & figures
python -m pytest tests/ -q               # 74 tests
```

Stage 2 writes to `outputs/transitions/`: nine CSV tables (transition matrices with and without intervals, prevalence, reversion, pathways, both Kaplan–Meier curves), six figures, and `report.txt` containing every number quoted here. All item lists, thresholds, cohort rules and interval bins live in `config/config.yaml`.

Reference. Stebbins GT, Goetz CG, Burn DJ, Jankovic J, Khoo TK, Tilley BC. How to identify tremor dominant and postural instability/gait difficulty groups with the Movement Disorder Society Unified Parkinson's Disease Rating Scale. *Movement Disorders* 2013;28(5):668–70.